

Mathematics I

Week 5: Graded Assignment Practice Session



Plan for this session

- How to join?
 - Join on webex - click on link sent to you
 - Join on pear deck - joinpd.com (enter code: see top right on webex)
 - Keep a notebook and pen ready for solving problems
- For every question - 5 to 15 minutes allotted
 - Question will be shown in a slide for solving
 - If you are done solving, enter your answer on joinpd.com
 - Presenter will provide a solution
 - Questions and discussion

Example Screenshots

Laptop/Desktop

The screenshot shows a presentation slide with a dark background. On the left, there is a large grey rectangular area labeled "Q1 (a)". To its right, the text reads: "Is this function even or odd or neither even nor odd?". In the top right corner, a small box contains the text "How to participate? joinpd.com code". At the bottom left, there is a small cartoon character and the text "Students choose an option". At the bottom right, it says "Pear Deck Interactive Slide Do not remove this bar".

Portion for Answering

Even

Odd

Neither even nor odd

Mobile

The screenshot shows the same presentation slide on a mobile device. The top status bar shows the time as 2:50 PM and various icons. The address bar shows "app.peardeck.com/studi". The slide content is the same as the laptop/desktop version. At the bottom, there is a green button labeled "Answer Question" which is circled in red. An arrow points from the text "Portion for Answering" to this button.

Portion for Answering

Answer Question

Test Problem

This is a problem to test if everything is working.

What is the chance that India will draw the first test against England? Give in percentage.



Students, enter a number!

Q1

How to participate?
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code: see above

Consider the function $f(x) = x^2 - 5x + 6$.

Plot the graph of $f(x)$.

What is the range of $f(x)$?

(a) $(-\infty, 0.25]$

(b) $[0.25, \infty)$

(c) $(-\infty, -0.25]$

(d) $[-0.25, \infty)$

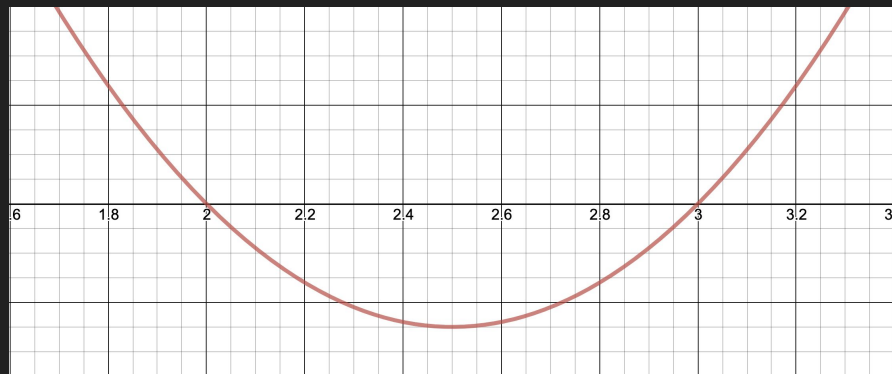


Students choose an option

Solution: Plotting a quadratic

How to participate?
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code: see above

- Factor $f(x)$
 $x^2 - 5x + 6 = (x - 2)(x - 3)$
- Range: x in $(-\infty, 2)$
 - $x - 2$: negative, $x - 3$: negative
 - $x^2 - 5x + 6$: positive
- Range: x in $(2, 3)$
 - $x - 2$: positive, $x - 3$: negative
 - $x^2 - 5x + 6$: negative
- Range: x in $(3, \infty)$
 - $x - 2$: positive, $x - 3$: positive
 - $x^2 - 5x + 6$: positive

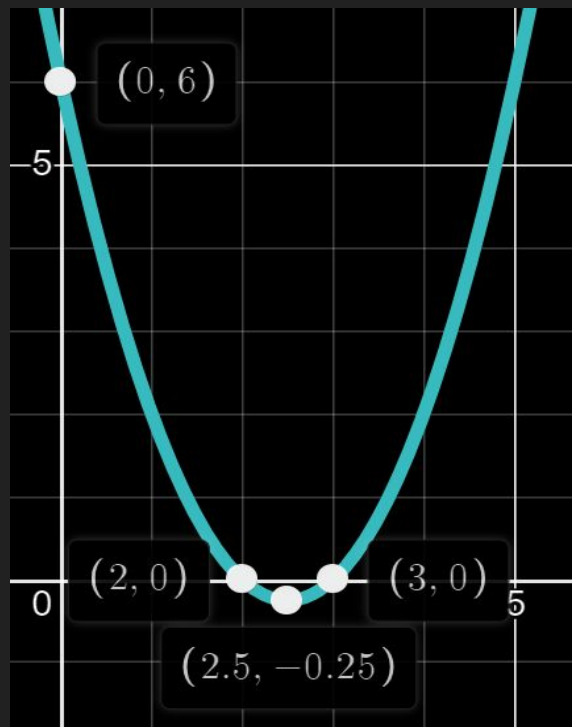


desmos

- $x \rightarrow -\infty$ and $x \rightarrow \infty$ for any polynomial
 - $\text{LeadingTerm}(-\infty)$ and $\text{LeadingTerm}(\infty)$
- Value at the bend for degree 2
 - $x_{\text{bend}} = -(\text{coeff of } x) / (2 \times \text{coeff of } x^2)$
 - Value at bend: $f(x_{\text{bend}})$

Solution: Finding range of a quadratic

How to participate?
joinpd.com
code: see above

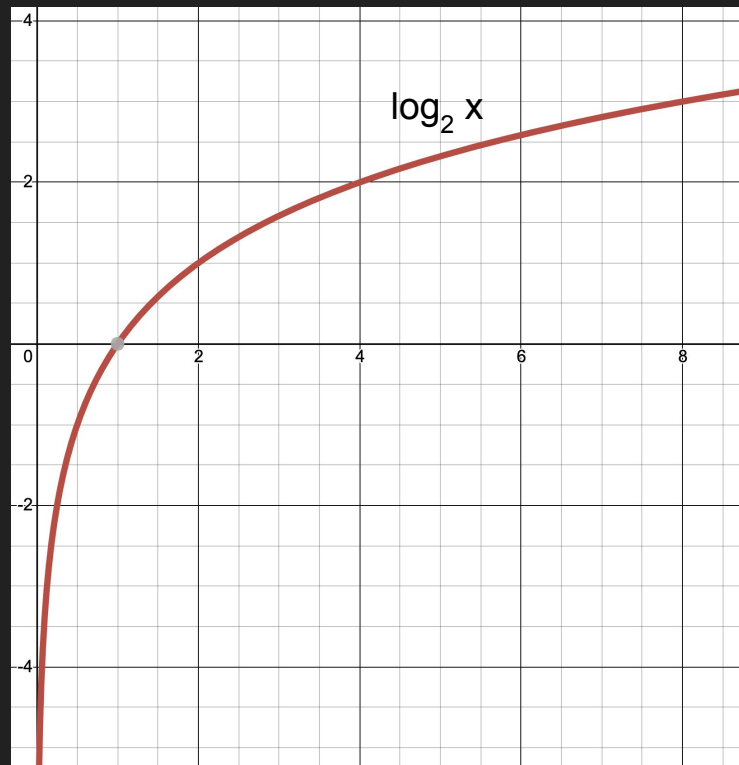


- $x \rightarrow -\infty$ and $x \rightarrow \infty$ for quadratic
 - $a = \text{LeadingTerm}(\infty) = \text{LeadingTerm}(-\infty)$
- Value at the bend for degree 2
 - $x_{\text{bend}} = -(\text{coeff of } x) / (2 \times \text{coeff of } x^2)$
 - Value at bend: $b = f(x_{\text{bend}})$
- Range: if $a = +\infty$, $[b, +\infty)$
if $a = -\infty$, $(-\infty, b]$

Answer: $[-0.25, \infty)$

Quick primer on $\log_a x$

- Assume $a > 1$ (if $a < 1$, $\log_a x = -\log_{1/a} x$)
- Plot of $\log_a x$ versus x
 - Only for $x > 0$
 - As x decreases to 0, $\log_a x$ decreases to $-\infty$
 - As x goes from 0 to 1, $\log x$ goes from $-\infty$ to 0
 - $\log_a 1 = 0$
 - As x increases to $+\infty$, $\log_a x$ increases to $+\infty$
 - As x goes from 1 to ∞ , $\log x$ goes from 0 to ∞
- Properties
 - $\log ab = \log a + \log b$
 - $\log a/b = \log a - \log b$
 - $\log a^b = b \log a$
 - One-to-one, invertible: inverse a^x



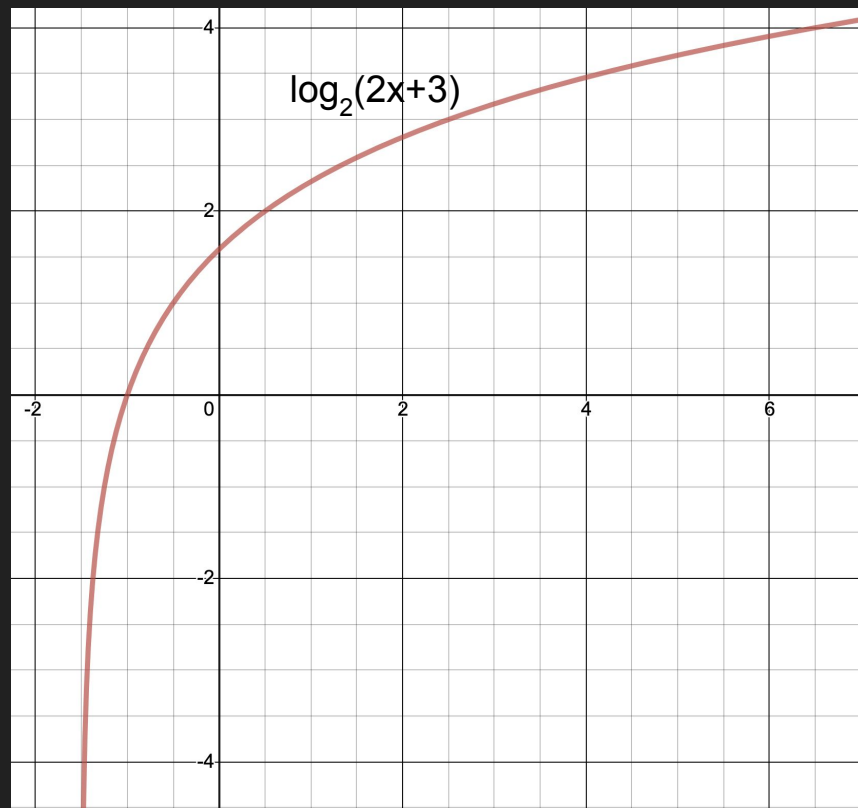
Plotting log of other functions

How to participate?
joinpd.com
code: see above

Consider the function $f(x)=\log_2(2x+3)$.

$$g(x) = 2x + 3$$

- When is $g(x) = 0$? $x = -3/2$
- When does $g(x) \rightarrow \infty$? $x \rightarrow \infty$
- For what x is $g(x) > 0$? x in $[-3/2, \infty)$
- For more points....
 - When is $g(x) = 1$? $f(-1) = 0$
 - When is $g(x) = 2$? $f(-1/2) = 1$
 - When is $g(x) = 4$? $f(1/2) = 2$
 - When is $g(x) = 8$? $f(5/2) = 3$



Q2

How to participate?
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code: see above

Consider the function $f(x) = \log_2(-36 + 20x - x^2)$.

This is defined whenever $-36 + 20x - x^2 > 0$
(usually not stated but assumed).

What is the range of $f(x)$?

- (a) $(-\infty, 64]$
- (b) $[6, \infty)$
- (c) $[0, \infty)$
- (d) $(-\infty, 6]$



Students choose an option

Solution: log of a function

How to participate?
joinpd.com
code: see above

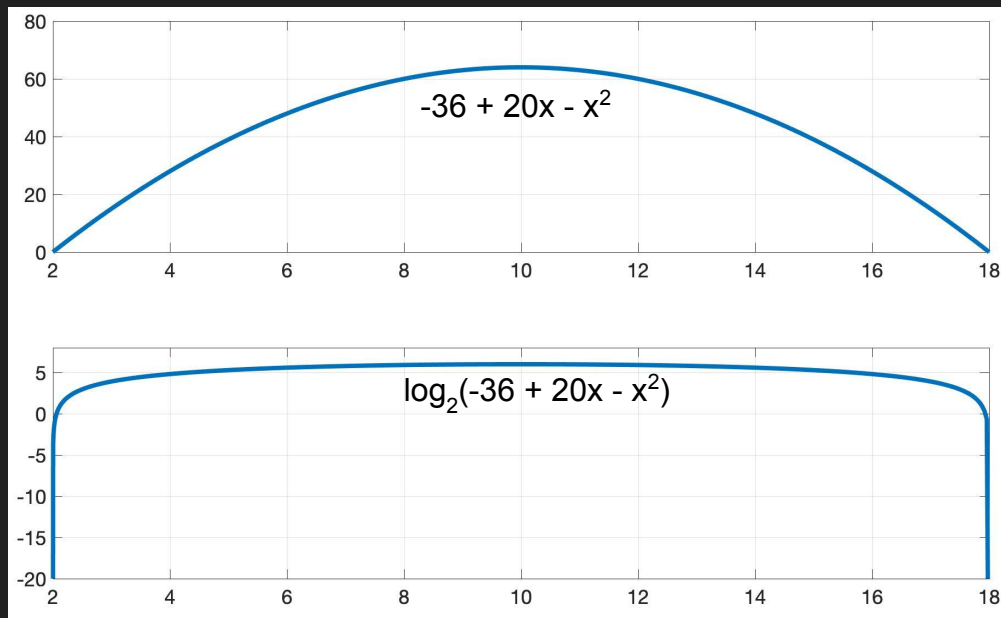
$$g(x) = -36 + 20x - x^2 = -(x-2)(x-18)$$

- $g(x) = 0$: $x = 2$ and 18
- $g(x) > 0$? x in $(2, 18)$
- $g(x) \rightarrow \infty$? No.
- Where is the peak? $x = 10$
 - Value at peak = 64

$$f(x) = \log(-36 + 20x - x^2)$$

- At $x = 2$ and $x = 18$, $f(x)$ goes to $-\infty$
- At $x = 10$, $f(x)$ peaks at
 $f(64) = \log_2 64 = 6$

Answer: $(-\infty, 6]$



Increasing/decreasing and log

- $\log x$ is one-to-one and increasing
 - Domain: $(0, \infty)$
 - Range: $(-\infty, \infty)$
- Taking log on both sides of an equation
 - If $x_1 = x_2$ and both x_1 and x_2 are in the domain of log, then $\log x_1 = \log x_2$
 - If $\log x_1 = \log x_2$, then $x_1 = x_2$
- Taking log on both sides of an inequality
 - If $x_1 < x_2$ and both x_1 and x_2 are in the domain of log, then $\log x_1 < \log x_2$
 - If $\log x_1 < \log x_2$, then $x_1 < x_2$
 - $<$ can be replaced with $\leq$ above

Q3

How to participate?
joinpd.com
code: see above

Consider a function $f(x) = \log_{10}(x^2 - x + 1)$.

Which of the following are true?

- (a) $f(x)$ is a one to one function on the interval $(0.5, \infty)$
- (b) $f(x)$ is strictly increasing function on the interval $(0.5, \infty)$
- (c) $f(x)$ is strictly decreasing function on the interval $(0.5, \infty)$
- (d) None of the above



Students, write your response!

Solution: log of another function

How to participate?
joinpd.com
code: see above

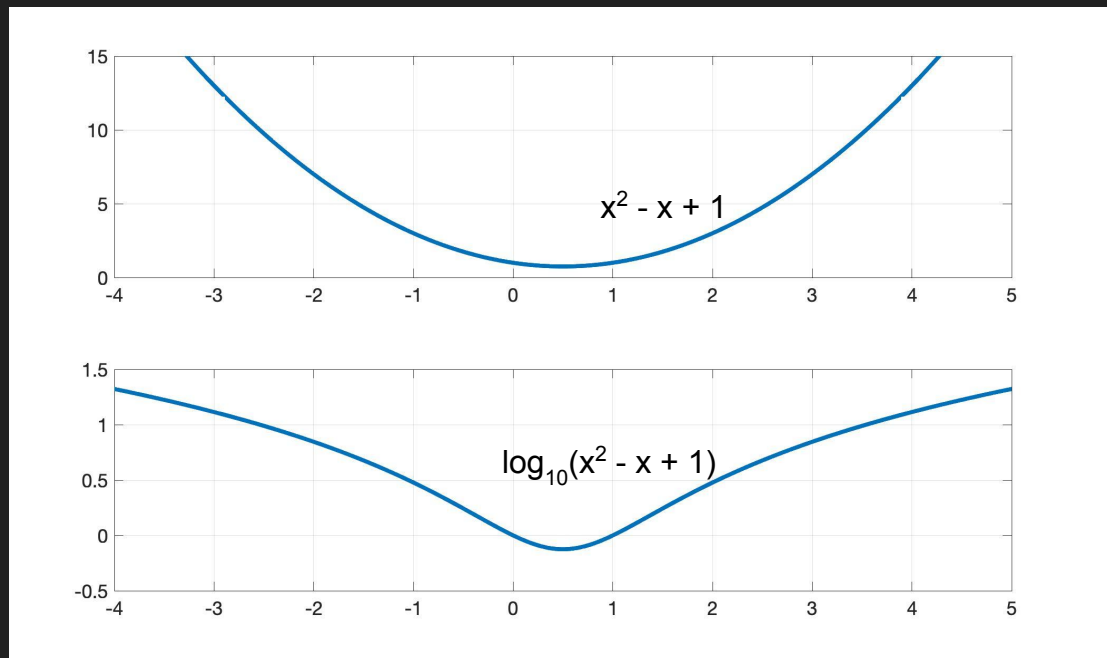
$$g(x) = x^2 - x + 1$$

- $g(x) = 0$? no real roots
- $g(x) \rightarrow \infty$? $x \rightarrow -\infty$ or ∞
- Where is the bend? $x = 1/2$
 - Value at bend = $3/4$
- $g(x) > 0$? For all x

$$f(x) = \log_{10}(x^2 - x + 1)$$

- At $x \rightarrow -\infty$ or ∞ , $f(x) \rightarrow \infty$
- At $x = 1/2$, $f(x)$ bends at $f(3/4) = \log_{10}(3/4)$

Answer: a, b



Solution: log of another function

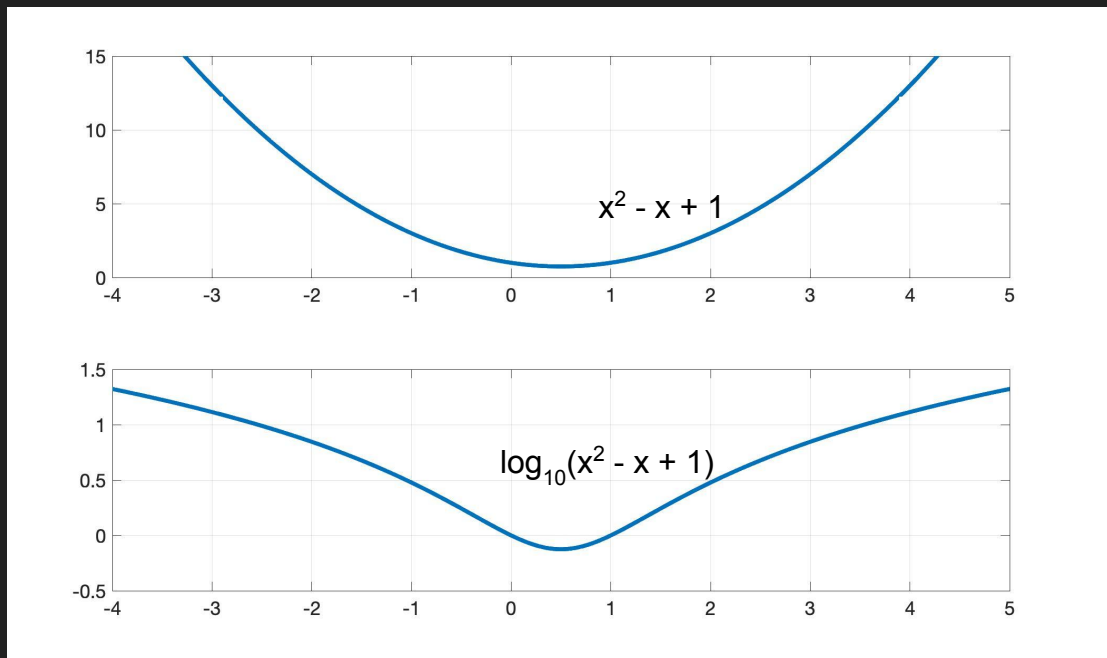
How to participate?
joinpd.com
code: see above

$$f(x) = \log_{10}(x^2 - x + 1)$$

- At $x \rightarrow -\infty$ or ∞ , $f(x) \rightarrow \infty$
- At $x = 1/2$, $f(x)$ peaks at $f(3/4) = \log_{10}(3/4)$

For what x , is $f(x) = 0$?

Hint: if $f(x) = 0$,
what should $x^2 - x + 1$ be?



Students, write your response!

Q4

How to participate?
joinpd.com
code: see above

Consider the function

$$f(x) = \ln (x / (5-6x)).$$

Hint: Find the domain and then use log properties.

Choose the correct option from the following

- (a) $f(x)$ is neither increasing nor decreasing in its domain
- (b) $f(x)$ is a decreasing function
- (c) $f(x)$ is an increasing function
- (d) None of the above



Students choose an option

Solution

$$g(x) = x / (5 - 6x)$$

- When is $g(x) = 0$? $x = 0$
- When does $g(x) \rightarrow \infty$? $x \rightarrow 5/6$
- For what x is $g(x) > 0$?
 $x > 0$ and $(5-6x > 0$ or $x < 5/6)$ (yes)
or
 $x < 0$ and $(5-6x < 0$ or $x > 5/6)$ (no)
So, $0 < x < 5/6$

Consider $x_1 < x_2$ and $x_1, x_2 \in (0, 5/6)$

$$-6x_1 > -6x_2 \Rightarrow 5 - 6x_1 > 5 - 6x_2$$

$$1/(5 - 6x_1) < 1/(5 - 6x_2)$$

$g(x)$: increasing in $(0, 5/6)$

Solution: Sketch this function

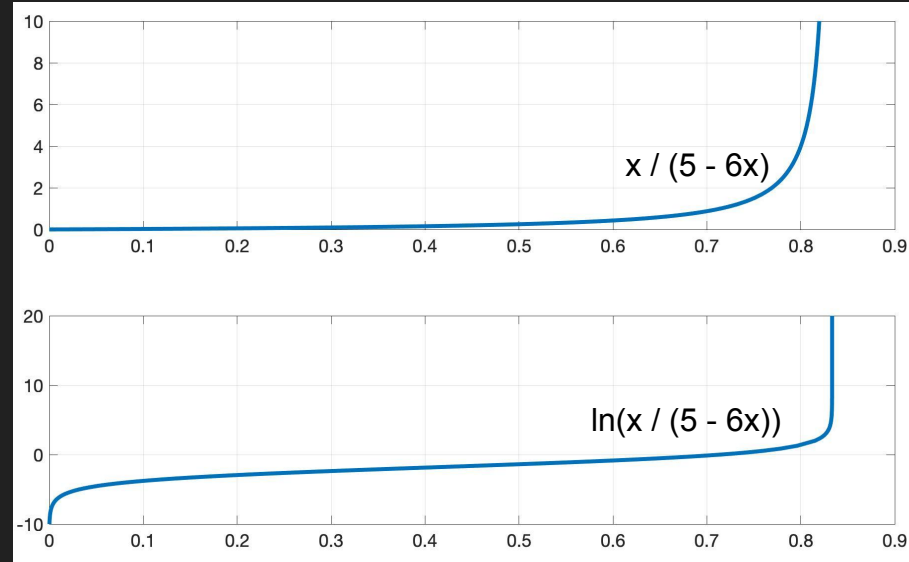
$$g(x) = x / (5 - 6x)$$

- When is $g(x) = 0$? $x = 0$
- When does $g(x) \rightarrow \infty$? $x \rightarrow 5/6$
- For what x is $g(x) > 0$?
 $x > 0$ and $(5-6x > 0$ or $x < 5/6)$ (yes, possible)
or
 $x < 0$ and $(5-6x < 0$ or $x > 5/6)$ (not possible)
So, $0 < x < 5/6$

$g(x)$: increases from $x = 0$ to $5/6$

$f(x) = \log g(x)$: increases from $x = 0$ to $5/6$

Answer: (c)



Q5

How to participate?
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code: see above

In a city, a rumour is spreading about the safety of corona vaccination. If

$$f(t) = N e^{-kt},$$

where $f(t)$ is the number of people who have not yet heard about the rumour after t days, k is a constant and N is the total population in the city. If the population of the city is 10000, and suppose 100 have heard the rumor after the first day.

After how many days, would half of the population have heard the rumor?



Students, enter a number!

Solution: Finding value of k

$$t = 1: f(t) = 10000 - 100 = 9900$$

$$9900 = 10000 e^{-k}$$

$$e^{-k} = 0.99$$

$$k = -\ln(0.99)$$

Solution

Given: $f(t) = 5000$, $k = -\ln(0.99)$, $N = 10000$

$$5000 = 10000 e^{\ln(0.99)t}$$

$$\ln(0.5) = \ln(0.99)t$$

$$t = \ln(0.5)/\ln(0.99) \approx 69 \text{ days}$$

Q6

How to participate?
joinpd.com
code: see above

Consider a functions $f(x) = (15e^x)/(e^x+1)$

Hint:

Check $f(x)$ is one-to-one function and then proceed.

Which of the following option is inverse of $f(x)$

(a) $f^{-1}(x) = \ln (y/(15-y))$

(b) $f^{-1}(x) = \ln ((15-y) / y)$

(c) $f^{-1}(x) = \ln ((15-y) / (15+y))$

(d) $f^{-1}(x) = 2 \ln (15-y)$



Students choose an option

Solution: Check whether $f(x)$ it is one-to-one

Use the property of one-to-one function, if $f(x_1) = f(x_2)$ then $x_1 = x_2$

$$(15e^{x_1})/(e^{x_1}+1) = (15e^{x_2})/(e^{x_2}+1)$$

$$\Rightarrow e^{x_1}(e^{x_2}+1) = e^{x_2}(e^{x_1}+1)$$

$$\Rightarrow e^{x_1+x_2} + e^{x_1} = e^{x_1+x_2} + e^{x_2}$$

$$\Rightarrow e^{x_1} = e^{x_2}$$

$$\Rightarrow x_1 = x_2$$

- Since $f(x)$ is one-to-one function, inverse exists.

Solution : Finding inverse of $f(x)$

- $f(x) = y = (15e^x)/(e^x+1)$, solve for x in terms of y gives f^{-1}

$$ye^x + y = 15e^x$$

$$15e^x - ye^x = y$$

$$e^x(15-y) = y$$

$$e^x = y/(15-y) \text{taking ln on both side}$$

$$x = \ln (y/(15-y)) = f^{-1}$$

- So, $f^{-1}(x) = \ln (x/(15-x))$

Option (a) is correct.

Q7

How to participate?
joinpd.com
code: see above

Given that $m = (8!)^{1/8}$ and $n = (9!)^{1/9}$,
where $x! = x(x-1)(x-2)(x-3).....2 \times 1$,
then choose the correct option.

Hint: $8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

$9! = 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

$9! = 9 \times 8!$

Take $\log_{8!} ()$.

Options:

- (a) $m = n$
- (b) $m < n$
- (c) $m > n$
- (d) None of above



Students choose an option

Solution

Option (b): How to check $m < n$?

$$(8!)^{1/8} < (9!)^{1/9}$$

$$\text{Only if } \log_{8!}(8!)^{1/8} < \log_{8!}(9!)^{1/9}$$

$$\text{Only if } 1/8 (\log_{8!}(8!)) < 1/9 (\log_{8!}(9!))$$

$$\text{Only if } 1/8 \times 1 < 1/9 \log_{8!}(8! \times 9)$$

$$\text{Only if } 9/8 < \log_{8!}(8!) + \log_{8!}(9)$$

$$\text{Only if } 9/8 < 1 + \log_{8!}(9)$$

$$\text{Only if } 1/8 < \log_{8!}(9), \text{ only if } 1 < 8 \log_{8!}(9)$$

$$\text{Only if } 1 < \log_{8!}(9^8), \text{ only if } 8! < 9^8, \text{ which is true.}$$

Thank You

Q4

How to participate?
joinpd.com
code: see above

Let $f(x)$ be a one-one function defined for the domain $D \subseteq \mathbb{R}$ and $g(x) = \log(f(x))$ defined for domain D , then $g(x)$ is:

Choose the correct option(s).

- (a) a one-one function.
- (b) a function but not one-one.
- (c) not a function.
- (d) could be one-one function.



Students choose an option

Solution

Step I: Use the property of one-one function

- For $x_1 \neq x_2$, where $x_1, x_2 \in D$
- $f(x_1) \neq f(x_2)$

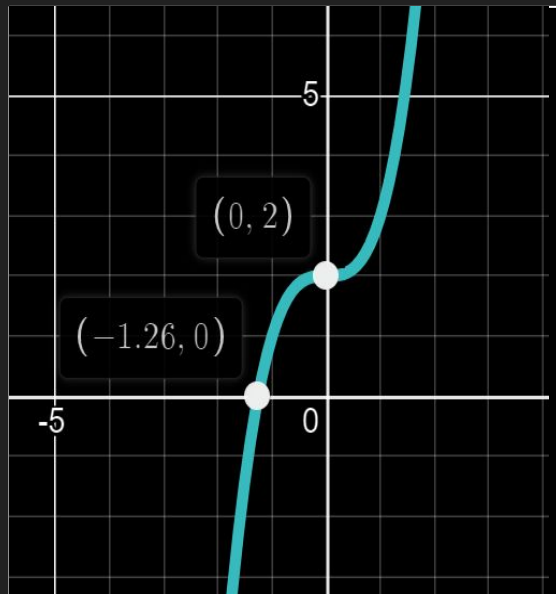
Step II: Use the property of log function

- As $\log(x)$ is an one-one function then $\log(x_1) \neq \log(x_2)$, for $x_1 \neq x_2$
- Similarly, $\log(f(x_1)) \neq \log(f(x_2))$, for $f(x_1) \neq f(x_2)$

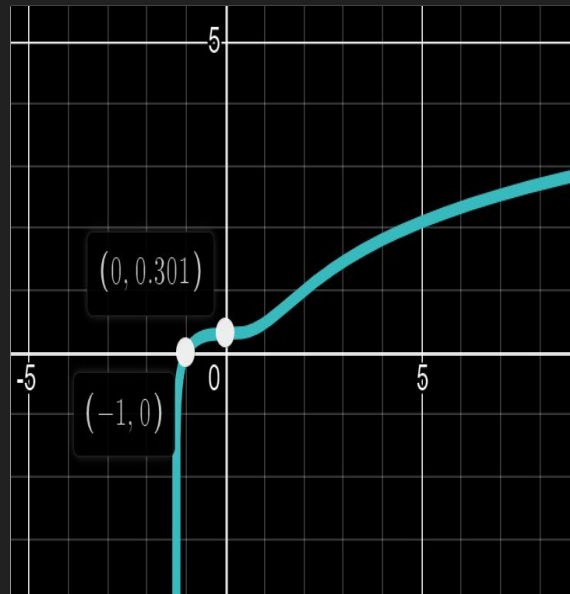
Thus option (a) is right

Example:

$$f(x) = x^3 + 2$$



$$g(x) = \log(f(x)) = \log(x^3 + 2)$$



Q5

How to participate?
joinpd.com
code: see above

Given that $m = (8!)^{1/8}$ and $n = (9!)^{1/9}$,
where $x! = x(x-1)(x-2)(x-3)..2.1$,
then choose the correct option.

Hint:

$$8! = 8.7.6.5.4.3.2.1$$

$$9! = 9.8.7.6.5.4.3.2.1$$

$$9! = 9.8!$$

Options:

- (a) $m = n$
- (b) $m < n$
- (c) $m > n$
- (d) None of above



Students choose an option

Solution

Option (a): $m = n$

$$(8!)^{1/8} = (9!)^{1/9}$$

$$\Rightarrow \log_{8!}(8!)^{1/8} = \log_{8!}(9!)^{1/9}$$

$$\Rightarrow 1/8 (\log_{8!}(8!)) = 1/9 (\log_{8!}(9!))$$

$$\Rightarrow 1/8 \times 1 = 1/9 \log_{8!}(8! \times 9)$$

$$\Rightarrow 9/8 = \log_{8!}(8!) + \log_{8!}(9)$$

$$\Rightarrow 9/8 = 1 + \log_{8!}(9)$$

$$\Rightarrow 1/8 = \log_{8!}(9)$$

$$\Rightarrow 1 = 8 \log_{8!}(9)$$

$$\Rightarrow 1 = \log_{8!}(9^8)$$

$$\Rightarrow 9^8 = 8!$$

Not possible case

Solution

Option (b): $m < n$

$$(8!)^{1/8} < (9!)^{1/9}$$

$$\Rightarrow \log_{8!}(8!)^{1/8} < \log_{8!}(9!)^{1/9}$$

$$\Rightarrow 1/8 (\log_{8!}(8!)) < 1/9 (\log_{8!}(9!))$$

$$\Rightarrow 1/8 \times 1 < 1/9 \log_{8!}(8! \times 9)$$

$$\Rightarrow 9/8 < \log_{8!}(8!) + \log_{8!}(9)$$

$$\Rightarrow 9/8 < 1 + \log_{8!}(9)$$

$$\Rightarrow 1/8 < \log_{8!}(9)$$

$$\Rightarrow 1 < 8 \log_{8!}(9)$$

$$\Rightarrow 1 < \log_{8!}(9^8)$$

$$\Rightarrow 8! < 9^8$$

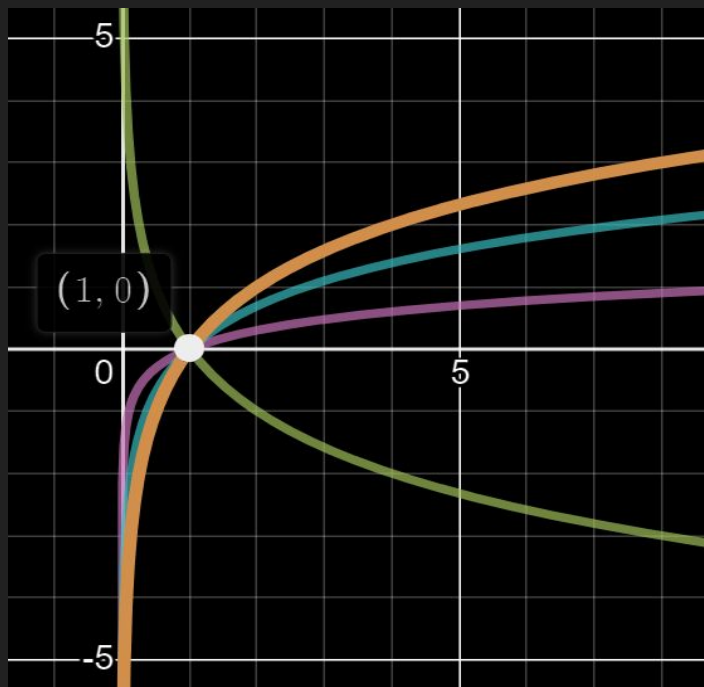
Possible case

Solution: Recall definitions and plots

How to participate?
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code: see above

Recall properties of $r(x) = \log_c x$

- For $c > 0$, $x \rightarrow 0$, $r(x) \rightarrow -\infty$ (Green, blue, black color)
- For $c < 0$, $x \rightarrow 0$, $r(x) \rightarrow \infty$ (Purple color)



Desmos plot:

Light green color: $r(x) = \log_{1/2} x$

Orange color: $r(x) = \log_2 x$

Pink color: $r(x) = \log x$

Blue color: $r(x) = \ln x$

Solution: Plot the quadratic

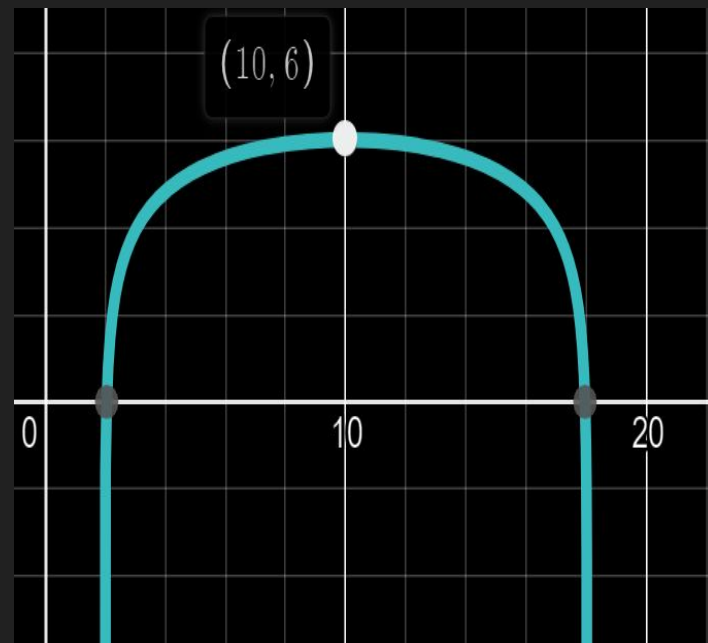
- $f(x) = q(h(x))$ is a composite function of $q(x)$ and $h(x)$ where $q(x) = \log_2 x$ and $h(x) = -36+20x-x^2$
- The maximum value of $f(x)$ can be obtained from the maximum value of $h(x) = -36+20x-x^2$.
 - Vertex of the quadratic equation $g(x) = ax^2+bx+c$ will be of the form $(-b/2a, g(-b/2a))$ where $g(-b/2a)$ gives the maximum or minimum value (If $a < 0$, then $g(x)$ will have maximum value ; if $a > 0$ then $g(x)$ will have minimum value).
 - For $h(x) = -36+20x-x^2$,
 - $a = -1$ (Note $a < 0$), $b = +20$, thus $-b/2a = 10$,
 $h(10)=64$;
 - $f(10) = \log_2(64) = 6$

Solution: To find the minimum value of $f(x)$

How to participate?
joinpd.com
code: see above

- Equate $h(x) = 0$ to find factors
 - $-36 + 20x - x^2 = 0$
 - $(x - 2)(-x + 18)$
- So $f(x) \rightarrow -\infty$ when $x \rightarrow 2$ and $x \rightarrow 18$
- Thus the range of $f(x)$ will be $(-\infty, 6]$

Answer: Option (a)



Desmos plot of $f(x)$

Solution

Step-1: Let $h(x) = x^2 - x + 1$ and $g(x) = \log_{10} x$

$$g(h(x)) = \log_{10}(h(x)) = \log_{10}(x^2 - x + 1) = f(x)$$

$$\text{So, } f(x) = g(h(x))$$

Step-2: Claim: $h(x) = x^2 - x + 1$ is increasing on $(0.5, \infty)$

$h(x) = x^2 - x + 1$ is quadratic function after comparison with general quadratic function $ax^2 + bx + c$, we got $a = 1 > 0$, $b = 1$, $c = 1$

It means graph of $h(x)$ open upwards and we will get the minimum value

Solution

How to participate?
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code: see above

(Step-2 continued):

at $x = -b/2a = 1/2 = 0.5$

For $x > 0.5$, function $h(x) = x^2 - x + 1$ is strictly increasing function

Step-3: Claim $h(x) > 0$ on $(0.5, \infty)$

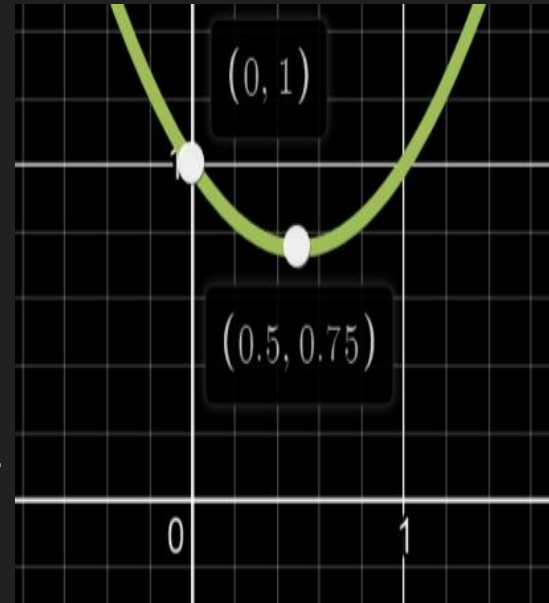
As we got $x = 0.5$, the function $h(x)$ has minimum value and

$$h(0.5) = (0.5)^2 - (0.5) + 1 = 0.75$$

and as in step-2 we proved $h(x)$ is strictly increasing function.

Hence, $h(x) > 0$ on $(0.5, \infty)$

Step-4: $g(x) = \log_{10} x$ and recall that logarithmic function is strictly increasing



Desmos plot
 $h(x) = x^2 - x + 1$

Solution

How to participate?
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code: see above

Step-5: From step-1, $f(x) = g(h(x)) = \log_{10}(x^2 - x + 1)$

From step- 2, $h(x)$ is strictly increasing function

From step-3, we are getting only positive values of $h(x)$ on $(0.5, \infty)$

From step-4, $\log_{10} x$ is strictly increasing function

So, here composition of two strictly functions will strictly increasing

So, $f(x)$ is strictly increasing function on $(0.5, \infty)$

Recall from video lecture, strictly increasing function is one to one Function. So, $f(x)$ is one to one function on $(0.5, \infty)$

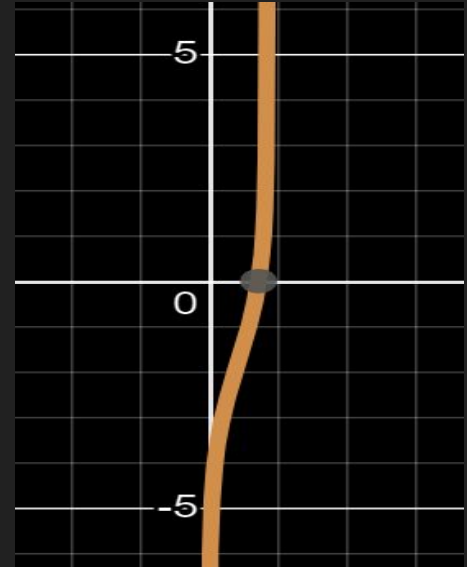
Answer- a,b

Solution: Finding domain of $f(x)$

Remember : $\log(x)$ function is defined for only $x > 0$

Let us say $f(x) = g(q(x)) = \ln(x/(5-6x))$ where $g(x) = \ln(x)$ and $q(x) = x/(5-6x)$

- Clearly $q(x) > 0$, so that $f(x)$ can be defined
 - Case 1:
 - $x > 0$ and $(5-6x) > 0 \Rightarrow x < \frac{5}{6}$
 - Thus, domain is $(0, \frac{5}{6})$
 - Case 2:
 - $x < 0$ and $(5-6x) < 0 \Rightarrow x > \frac{5}{6}$
 - No domain in this case as there is no intersecting points
 - Thus, overall the domain of $f(x)$ is $(0, \frac{5}{6})$



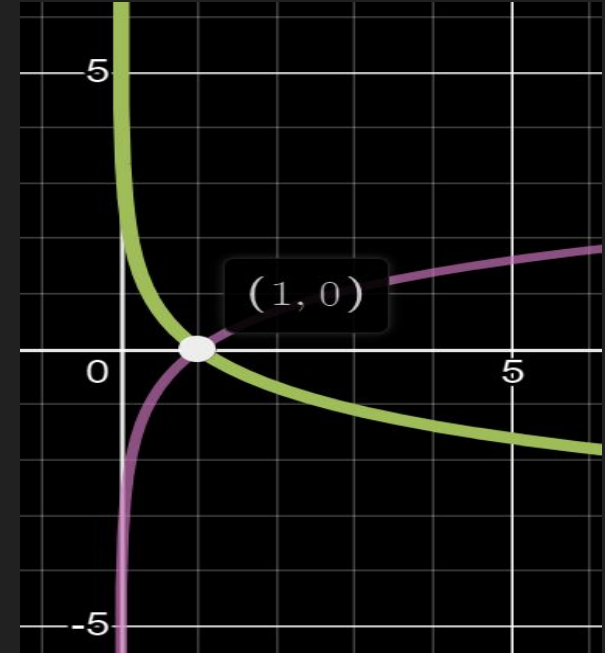
Desmos plot $f(x) = \ln(x/(5-6x))$

Solution : To check nature of $f(x)$ as per options

- We know that $\log_a x$ is increasing when $a > 1$ and decreasing when $a < 1$, therefore natural log (i.e., $\ln$) is increasing type function

Note:

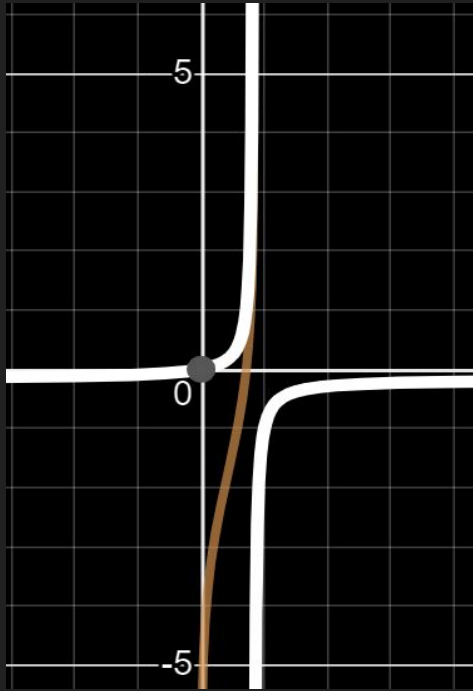
- $\ln$ of increasing function will be increasing
 - Example: $\ln(q(x))$ where $q(x) = x$ (see purple color in desmos plot)
- $\ln$ of decreasing function will be decreasing
 - Example: $\ln(q(x))$ where $q(x) = 1/x$ (see red color in desmos plot)



Desmos plot:
Green color: $\log_e(1/x)$
Pink color: $\log_e(x)$

Students, write your response!

Solution : To check nature of $f(x)$ as per options



Desmos plot

White color: $h(x) = x/(5-6x)$

Orange color: $f(x) = \ln(x/(5-6x))$

- Check whether $h(x) = x/(5-6x)$ is whether increasing or decreasing or neither in domain $(0, \frac{5}{6})$.
 - Consider $x_1 < x_2$ and $x_1, x_2 \in (0, \frac{5}{6})$
 - $-6x_1 > 6x_2$ also $5-6x_1 > 5+6x_2$
 - $1/(5-6x_1) < 1/(5+6x_2)$ Note ($x_1 < x_2$)
Therefore $x_1/(5-6x_1) < x_2/(5+6x_2)$ thus it is increasing type function and therefore $f(x)$ is increasing type function in domain $(0, \frac{5}{6})$.

Option (c) is right